

Extraction

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Extraction is the separation of a substance from a mixture by means of a solvent that preferentially dissolves that substance.

If the substance is extracted from a solid phase, the process is called solid-liquid extraction, as in the isolation of caffeine from tea leaves by means of hot water. Extraction of a substance from a liquid phase is called liquid-liquid extraction. The most common applications of this technique are:

- a) The recovery of an organic product from a reaction mixture containing excess unreacted materials and by-products
- b) Isolation of an organic substance from its natural source, such as a plant

Liquid-liquid extraction:-

This is the most common type of extraction. It involves shaking the liquid mixture with an immiscible solvent which preferentially dissolves the desired compound. On standing, the two immiscible phases (usually organic and aqueous) form two separate layers (upper and lower) that can be separated by means of a separatory funnel

The various solutes in the mixture distribute themselves between the organic and aqueous phases according to their relative solubilities in each solvent. At equilibrium, the ratio of the concentration (C) or solubility (S) of the substance in the organic phase (C_0 or S_0) to that in the aqueous phase (C_w or S_w) is called the distribution coefficient (K_D)

$$K_D = C_0/C_w = S_0/S_w$$

A large distribution coefficient implies that the compound is much more soluble in the organic phase than in the aqueous one and, in this case a single extraction suffices to remove the desired compound from the mixture. When K_D is small, it means that the compound distributes itself more evenly in both phases, so that repeated extractions are required to recover such a compound from the aqueous mixture.

In general, it is more efficient to divide the total volume of extracting solvent over several extractions than to use the whole volume in a single extraction.

General experimental considerations:-

Choice of solvent. A solvent used for extraction should have the following characteristics:

1. immiscible with the liquid in which the solute is present
2. readily dissolve the solute to be extracted
3. extract little or none of the impurities and other compounds present in the mixture
4. non flammable, non toxic, cheap and easily removable from solute after extraction(i.e. volatile)

salting-out. Extraction of organic compounds from aqueous mixture is usually improved by saturating the aqueous phase with a salt such as NaCl or Na₂CO₃. this phenomenon is called salting-out and has the following effects:

- 1- decreases the solubility of organic compounds in the saturated aqueous phase
- 2- decreases the solubility of the organic and aqueous phases in each other, thus improving their separation. This is particularly useful in breaking up emulsions

Emulsions:- in certain cases, the two immiscible phases do not separate clearly into two distinct layers, instead they form an emulsion which once formed is usually difficult to break. It is therefore advisable to prevent the formation of emulsions during extraction. This is best achieved by avoiding vigorous shaking of the layers whenever an emulsion is expected to form

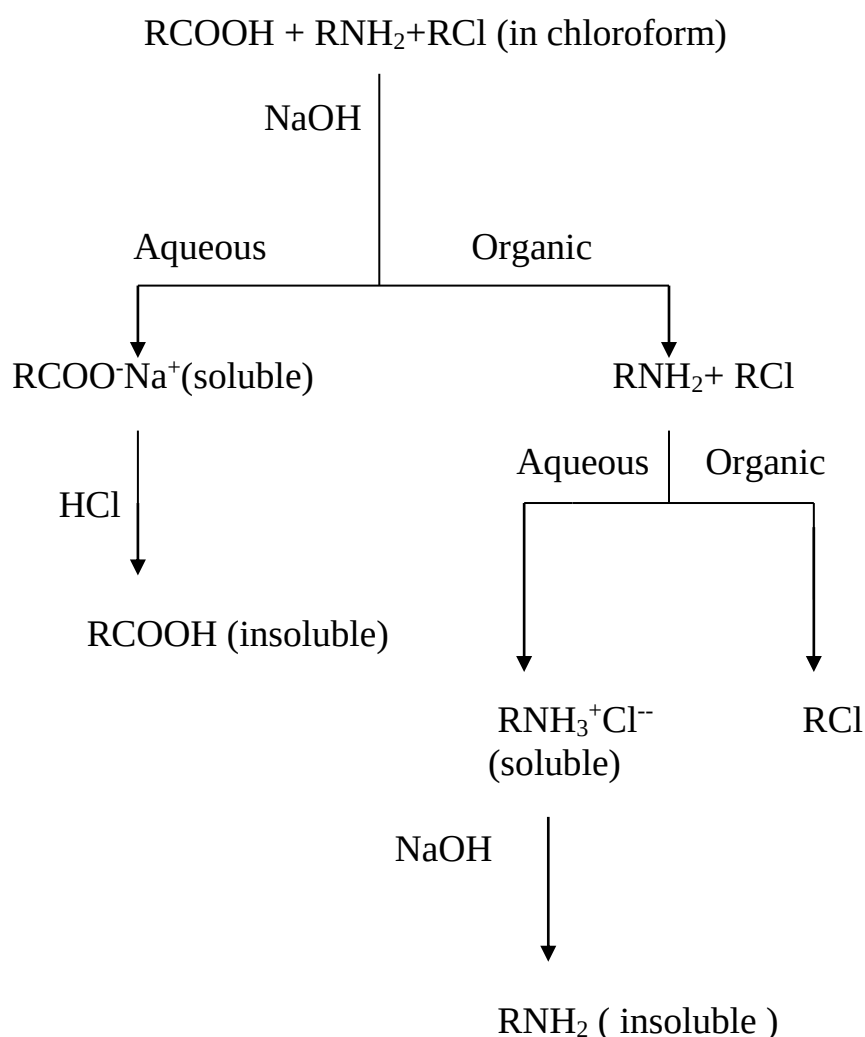
If an emulsion still forms one can often break it by:

- stirring the emulsified layer gently with a glass rod
- saturating the emulsified layer with a salt
- centrifugation

Drying agents:- the organic phase often shows turbidity due to the presence of traces of water from the aqueous phase. Anhydrous CaCl₂, MgSO₄, or Na₂SO₄ may be used as drying agents which absorb the traces of water present in the organic phase. When dry, the organic phase becomes clear.

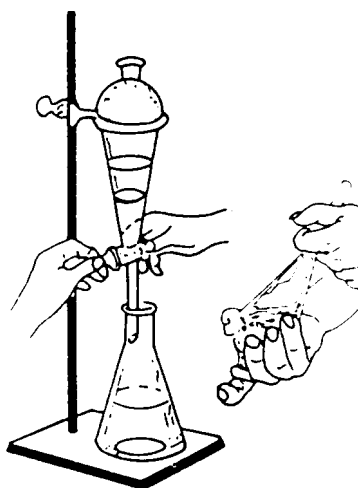
Acid-base Extraction

Mixtures of organic acids and bases are commonly separated by acid-base extraction. Such compounds are converted to their salts by treatment with acid or base. Unlike the original compounds, the corresponding salts are usually soluble in water, thus enabling their transfer from the organic phase to the aqueous layer. After separation of the layers, the organic acid or base is recovered by neutralization of the aqueous layer. Since the acid or base is insoluble in water it precipitates out, and is collected by filtration. A flow diagram for such a separation is shown in the following figure:



Apparatus and Procedure:-

The set-up for an extraction using the separatory funnel is shown in the following figure.



The stopcock and stopper should be greased before use. The separatory funnel is held upside down in both hands, such that the stopper is firmly held in the left hand, while the right hand controls the stopcock.

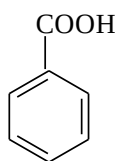
The funnel is shaken gently at the beginning and vented periodically through the stopcock to release excess pressure which build up inside.

Once there is no more pressure build-up in the funnel, shaking may be vigorous to ensure good extraction

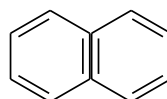
The funnel is placed in an iron ring, the stopper removed, and the two layers allowed to separate. The lower layer is drained through the stopcock, while the upper layer is poured through the top of the funnel into a separate container

Procedure:-

A mixture of Benzoic acid and Naphthalene will be separated into its components by means of extraction.



Benzoic acid



Naphthalene

- Obtain from your instructor a 2 g sample of the two-component mixture, dissolve it in 25 ml chloroform, & pour the solution into a 100 ml separatory funnel
- To extract the Benzoic acid from the mixture shake the chloroform solution with 20 ml of 10% NaOH. Draw off the chloroform layer (lower layer) into a beaker, pour the upper layer into a flask.
- Return the lower layer to the separatory funnel and repeat extraction of the chloroform layer with another 20 ml of 10% NaOH solution followed by 10 ml of water
- Combine the three aqueous extracts and cool in an ice-bath
- Neutralize the cold aqueous phase by adding conc. HCl until the solution is acid to blue litmus paper
- Collect the precipitated benzoic acid by suction filtration and wash with a little ice water
- Dry the product and determine its weight
- Pour the remaining chloroform solution into a small flask and add enough anhydrous CaCl_2 to remove any traces of water
- Decant the dry chloroform solution into a small weighed beaker and evaporate the solvent in the fume hood. Determine the weight of your product

Report sheet of Extraction

Student name

Mass of recovered Benzoic acid:

Mass of recovered Naphthalene

Questions:-

- 1) Why should the stopper be removed from the separatory funnel before liquid can be withdrawn through the stopcock?

- 2) What are the properties of a suitable solvent for the extraction of an organic solute from an aqueous solution?

- 3) What effect does partial miscibility of the two solvents have on the efficiency of the extraction ?

- 4) The distribution coefficient K_D ($C_{\text{ether}}/ C_{\text{water}}$) , for an organic substance X at room temperature is 10. what relative volumes of ether to water should be used for the extraction of 90% of X from a water solution in a single extraction?